1 Hydrogen Found in water, living organisms, hydrocarbons, plastics, fuel cells											**2 Helium** Occurs in the air, balloons, used as cooling medium in the Large Hadron Collider
3 Lithium Occurs in igneous rocks and electrical batteries	**4 Beryllium** In beryl, emerald and other minerals		**5 Boron** Found in volcanic water, used in Pyrex glass	**6 Carbon** In diamond, all living organisms and derivatives inc hydrocarbons	**7 Nitrogen** Occurs in air, proteins in all organisms, used in nitrate fertilisers	**8 Oxygen** Occurs in air, water all organisms, most metal ores and rust	**9 Fluorine** In some minerals, fluoridated water & toothpaste and Teflon	**10 Neon** Found in air and red advertising signs
11 Sodium In sea salt, living organisms, glass and soap	**12 Magnesium** In sea salt, fireworks and indigestion cures		**13 Aluminium** Occurs in kitchen foil, some drinks cans and aluminum oxide abrasive	**14 Silicon** Occurs in sand, clay and products e.g. glass, added to steel and transistors	**15 Phosphorous** Occurs in urine, bone and phosphate fertilisers	**16 Sulfur** Found in many minerals, with oil deposits, put in black rubber	**17 Chlorine** Occurs in sea salt, tap water, PVC plastic and antiseptics	**18 Argon** Occurs in air, put in light bulbs to protect the hot filament
19 Potassium In sea salt, fertilisers and living organisms	**20 Calcium** In bones, shells and hence limestone chalk, marble		**31 Gallium** In low melting point alloys e.g. automatic fire extinguishers and fuses	**32 Germanium** In fluorescent lamps, important semi-conductor in LEDs and DVD players	**33 Arsenic** To harden shot, solid-state devices e.g.in transistors	**34 Selenium** Occurs in solar cells, in stainless steel and in photocopiers	**35 Bromine** Occurs in sea water, dyes & flame retardants and petrol anti-knock	**36 Krypton** Occurs in air, used in some photographic flash lamps
37 Rubidium Occurs in some minerals	**38 Strontium** Occurs in some mineral and colour TV screens		**49 Indium** In transistors, thermistors and indium batteries in watches	**50 Tin** Used in food cans, found in pewter and bronze	**51 Antimony** In lead alloys to improve strength, infra-red detectors e.g. security lights	**52 Tellurium** In stainless steel, semi-conductors e.g. LEDs, DVD players	**53 Iodine** Occurs in sea water, photographic chemicals and dyes	**54 Xenon** Occurs in air, gives blue light so used in strobe lights
55 Caesium No common use (but used in atomic clocks)	**56 Barium** Occurs in green fireworks and preparation for gut X-rays		**81 Thallium** Used as rat poison	**82 Lead** Formerly used for water pipes, car batteries, lead crystal	**83 Bismuth** Used in low melting alloys e.g. in automatic fire extinguishers and fuses	**84 Polonium** Occurs rarely in nature and of very restricted use	**85 Astatine** Is not found in nature, no known use	
87 Francium Rare in nature, no known use	**88 Radium** A by-product of uranium refining, formerly used for luminous paint							

d-block (see next page)

p-block


s-block

21 Scandium Used in research. Alloy with aluminium	**22 Titanium** Used for strong steel. Oxide is the white pigment in paint	**23 Vanadium** Occurs in many minerals, added to make very tough steel	**24 Chromium** The red of rubies. Used in red, green yellow pigments	**25 Manganese** In many minerals. Found in magnetic steel	**26 Iron** In many minerals, found in red blood cells and in steel	**27 Cobalt** Used to cover other metals in blue glazes	**28 Nickel** Occurs in silver coins and gives glass a green colour	**29 Copper** Used in electrical cables and non-silver coins	**30 Zinc** In brass, stops iron rusting. In luminous dials and fluorescent lights
39 Ytttrium Used in the red of TV tube, oxide added to strengthen camera lenses	**40 Zirconium** Found in many minerals, used for furnace linings	**41 Niobium** Used to make unreactive steel e.g. industrial pipes	**42 Molyb-denum** Used to make tough steels for car engines	**43 Technetium** Stops steel rusting, used as a radioactive 'tracer'	**44 Ruthenium** Hardens electrical contacts	**45 Rhodium** Stops corrosion, makes very good mirrors in optical instruments	**46 Palladium** Used to stiffen metal in dental work	**47 Silver** Used for jewellery, photographs and tooth fillings	**48 Cadmium** In recharge-able batteries, in old colour TV tubes, yellow road paint
57 Lanthanum Occurs in carbon lights (studio and cinema lights)	**72 Hafnium** In incan-descent (filament) light bulbs	**73 Tantalum** In replacement parts for surgery, and in strip lights	**74 Tungsten** Used in the filaments of light bulbs and in drill bits	**75 Rhenium** In electrical contacts and mass spectro-graph filaments	**76 Osmium** In electrical contacts and instrument pivots	**77 Iridium** Used to make very hard pen tips e.g. in biro pens	**78 Platinum** Used in catalytic converters on car exhaust systems	**79 Gold** Used in jewellery and dental structures	**80 Mercury** Occurs in some ther-mometers

d-block

Teaching notes

This could be used in a variety of ways.

- Give students the information sheets and ask them to spot as many of the elements as they can at home or in school. You could provide them with a map (or get them to draw one of their home) and ask them to mark on it the atomic number of the element that they have spotted.

- Use the sheet as a quick quiz game. Give each group a copy of the information sheet. Then read out the information for an element, without revealing which on it is. When they find the element they should write it a mini whiteboard and hold it up. Award points for the first correct answer.

- Ask students to look at a group (e.g. the halogens, alkali metals or noble gases) and ask if they can spot any similarities within the group in terms of where they are found and their uses.

- Use as a starting point for further research into one of the elements or a group.

  The RSC's Periodic table is an excellent starting point for further research activities: www.rsc.org/periodic-table